Introduction to R and Python for Data Science

A Student Guide for Grading, Coding Style, and Project Milestones

Mahmoud Harding

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Preface

This resource is intended for students in *Introduction to R and Python for Data Science*. It provides materials and guidance to support understanding the grading rubrics, coding style, and project milestones.

Introduction

Purpose

This resource is intended for students in *Introduction to R and Python for Data Science*. It provides materials and guidance to support success throughout the course, including information about grading rubrics, coding style expectations, assignment requirements, and project milestones.

The guide is designed to help students understand how coursework will be evaluated and how to develop clear, readable, and well-documented code in both R and Python. It also outlines expectations for visualizations, written interpretations, and project milestones.

R Project Folder

Setting up Your RProject Folder

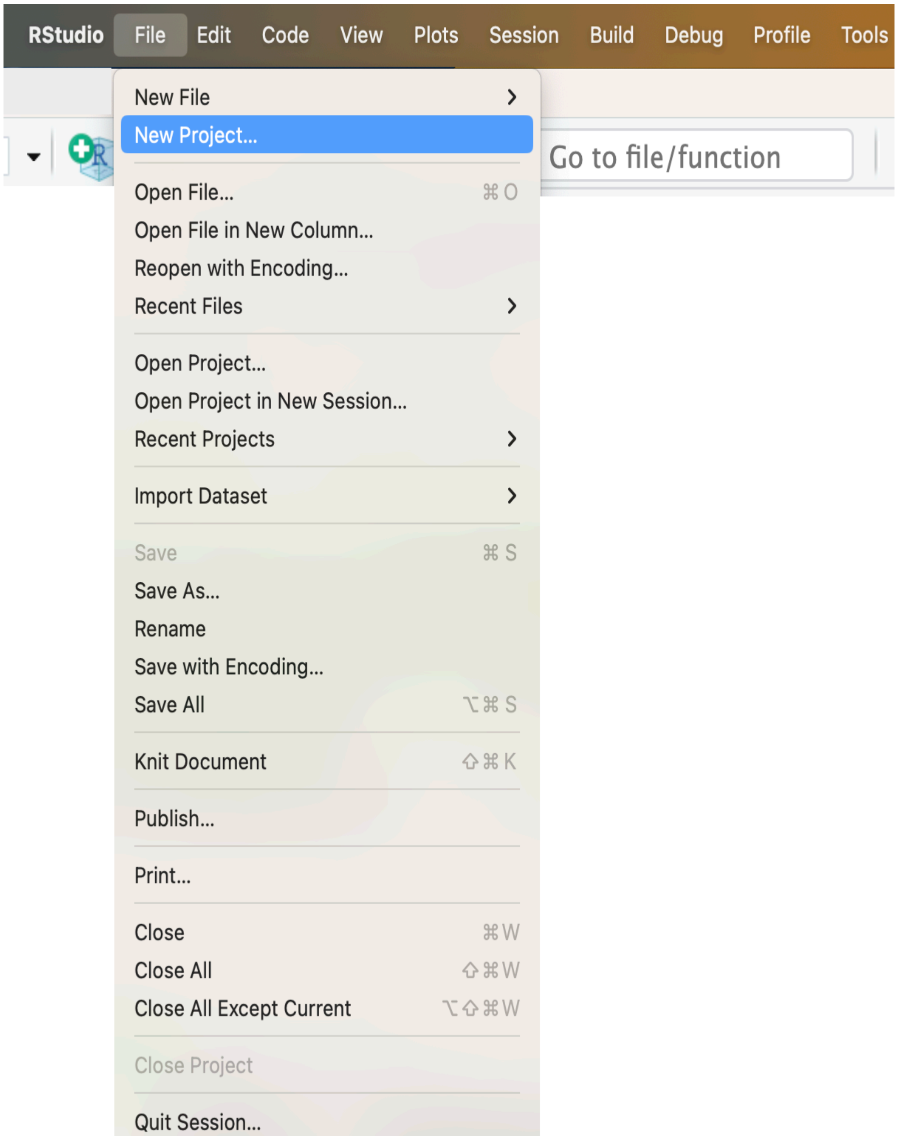
Follow these steps to:

- Create an Rproject folder named `dsc201`.
- Create a data folder inside the RProject folder.
- Download a dataset the `us_farmers_markets.csv` into the data folder.
- Download the `ds201_001-002_a00.Rmd` RMarkdown file into the RProject folder.

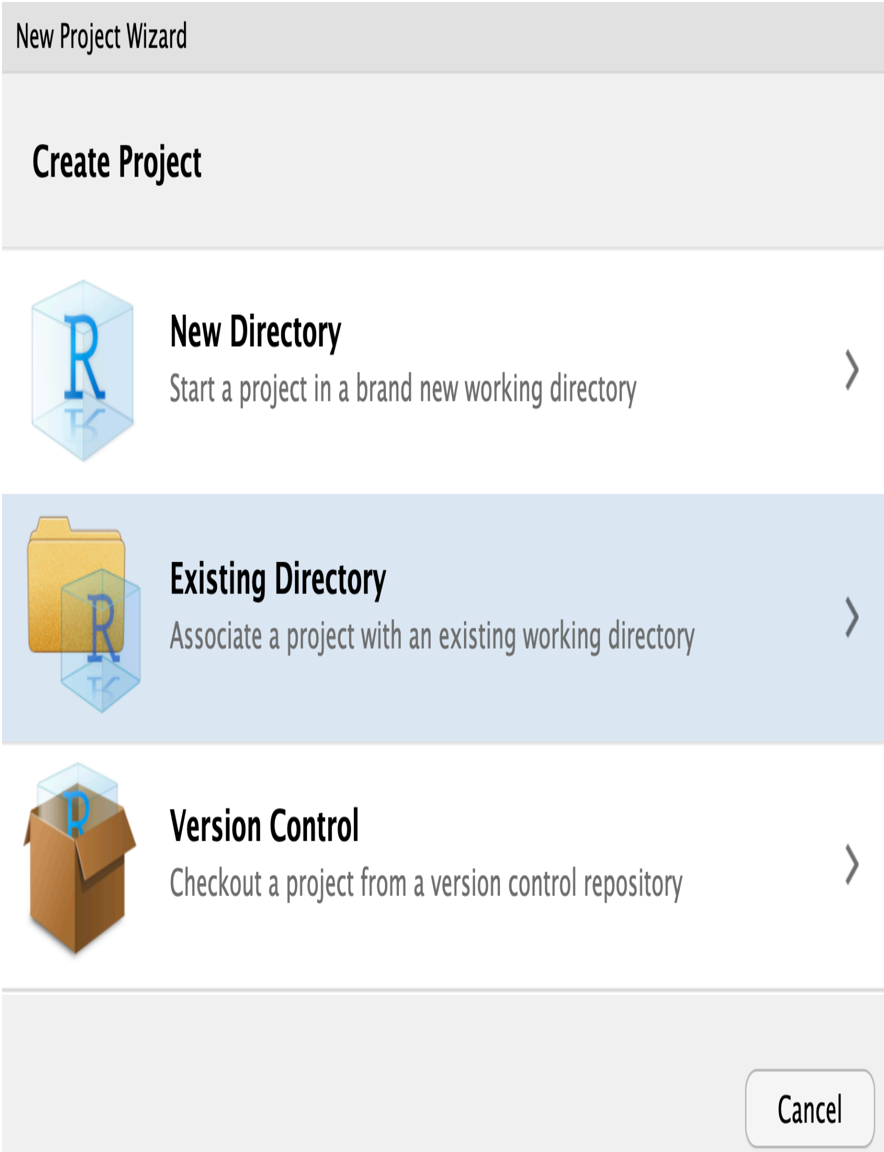
Note: Following these steps helps avoid file path errors and follows best practices from Martin Chan's RStudio Projects and Working Directories: A Beginner's Guide.

Step 1: Create a new RStudio project

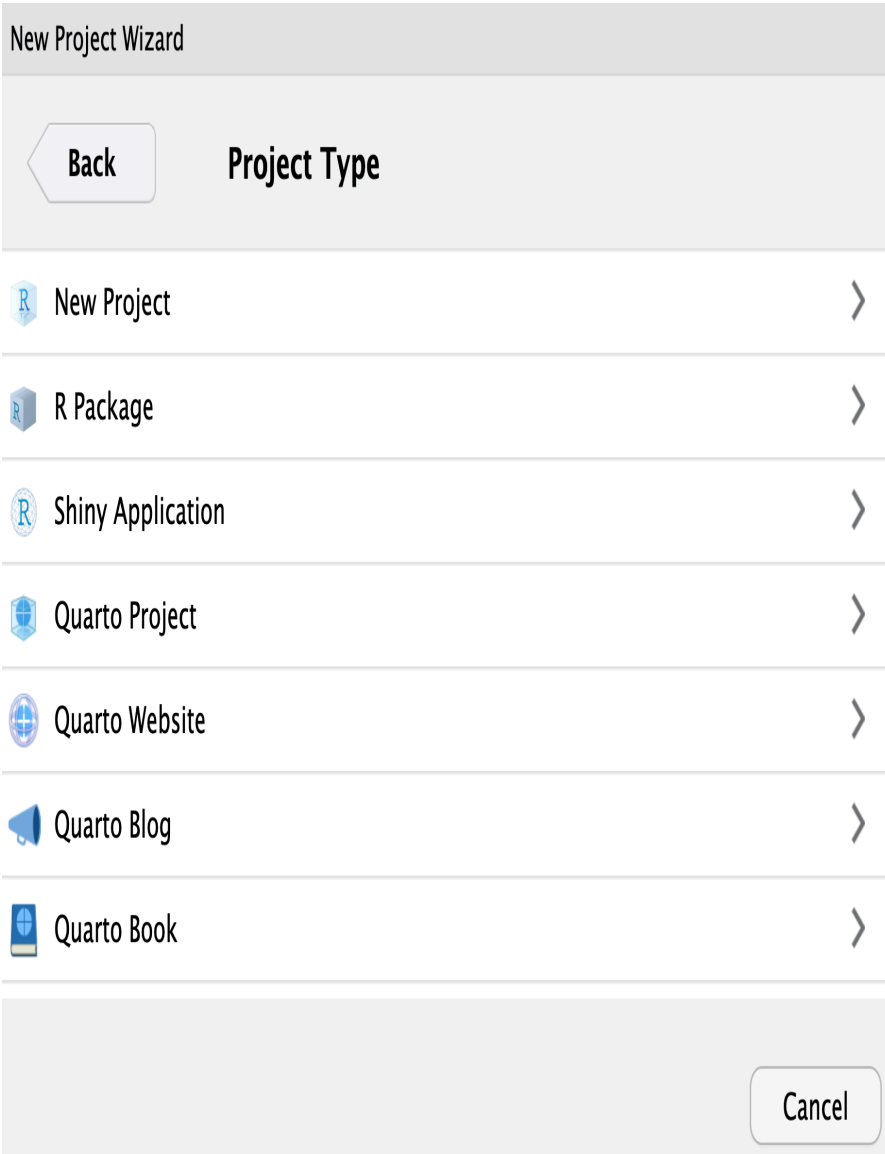
1. Open RStudio.
2. Go to **File** → **New Project...**



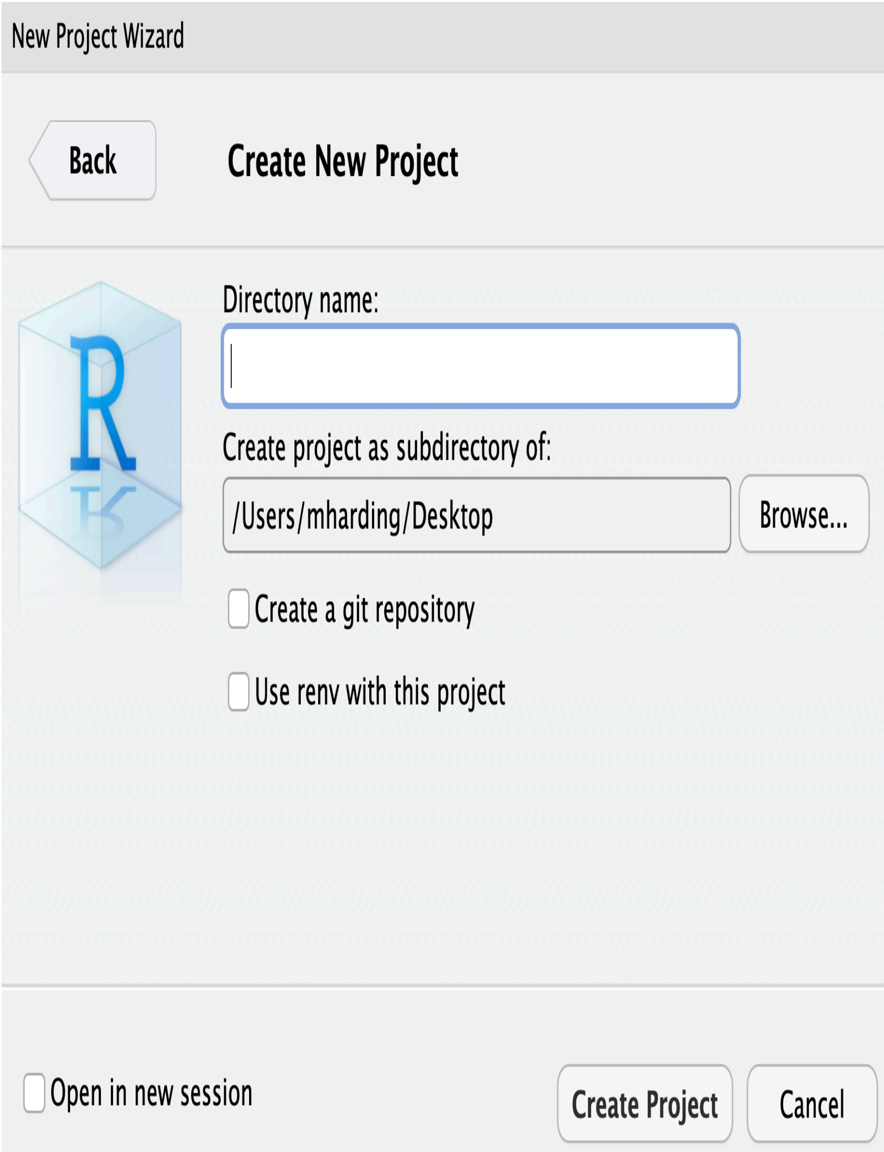
3. Select **New Directory** → **New Project**.



4. **Select New Project.**



5. In **Directory name:**, enter dsc201 (all lowercase)
6. In Create project as subdirectory of:, select Browse, then choose a location on your computer (e.g., Documents, Desktop, etc).



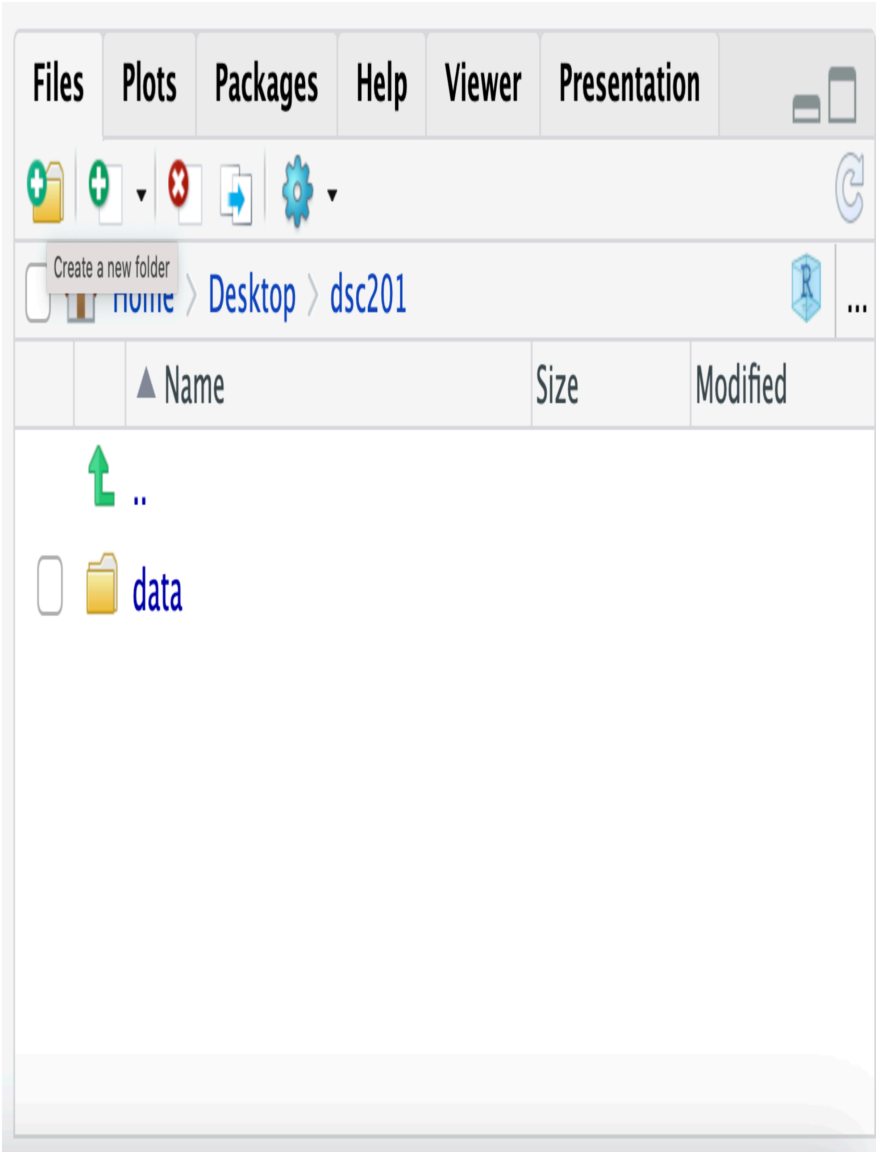
7. Click **Create Project**.

Note: Opening the `dsc201.Rproj` file will automatically set your working directory to the project's root folder.

Step 2: Create the data folder

Important: Name the folder `data` **(all lowercase)**.

8. In RStudio's **Files** pane (bottom right), click the **New Folder** icon.



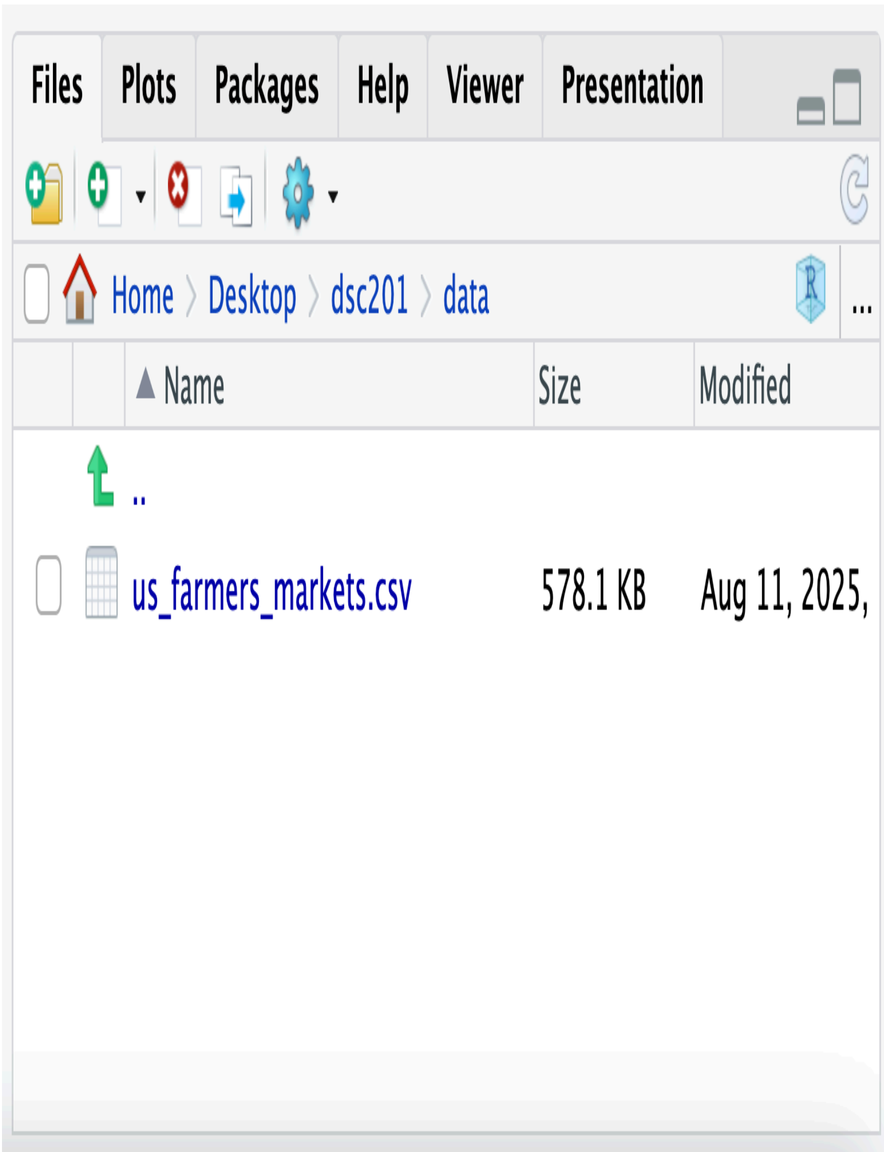
9. In **Please enter the new folder name**, enter data.
10. Click **OK**.

Important: Name the folder `data` **(all lowercase)**. Using all lowercase avoids issues on operating systems where `Data` and `data` are treated as different folders.

Step 3: Download the `us_farmers_markets.csv` data file

11. Download the `us_farmers_markets.csv` dataset from the course Moodle page.

12. Locate it in your browser's **Downloads** folder then move it manually into your `dsc201/data` RProject folder.
13. Confirm the file is in the right place by checking in RStudio's **Files** pane. You should see it in `data/us_farmers_markets.csv`.

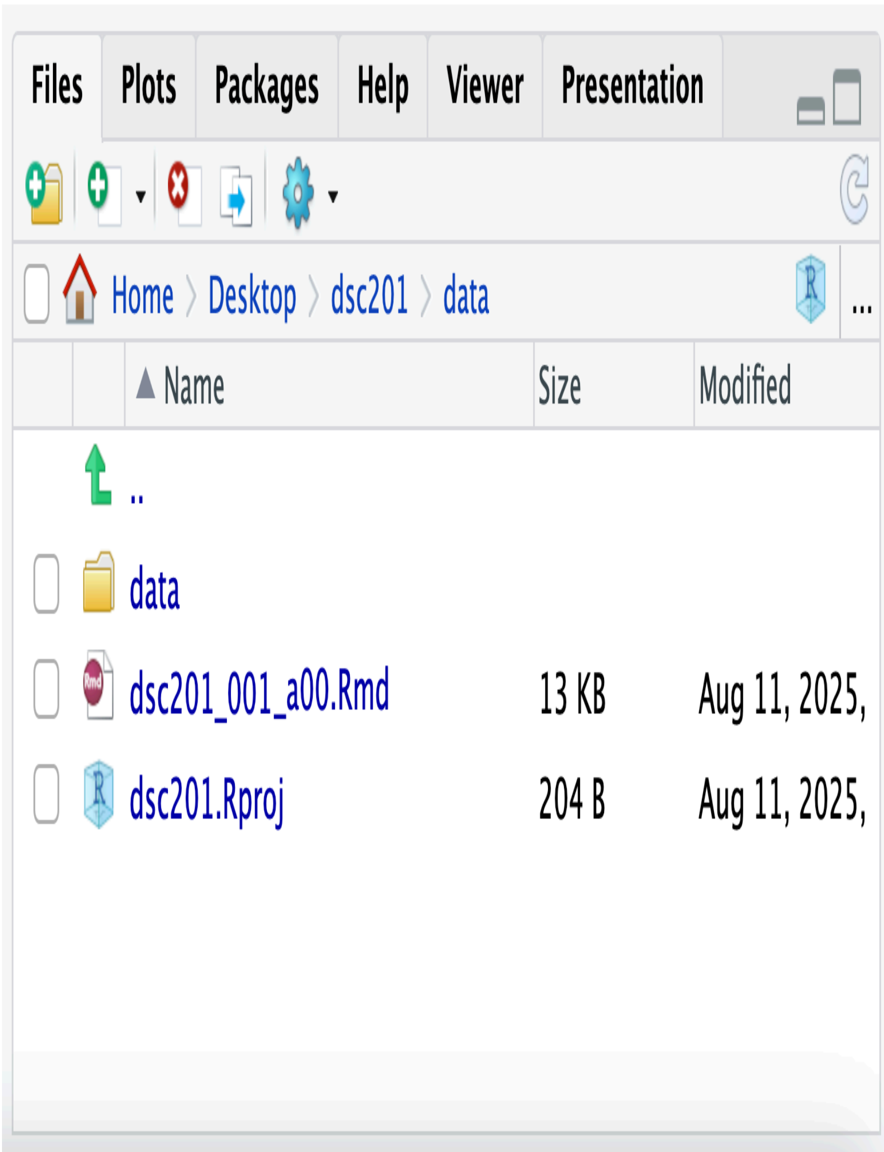


Step 4: Download the `.Rmd` file

14. Download the `dsc201_001_a00.Rmd` RMarkdown file from the course Moodle page.

15. Locate it in your browser's **Downloads** folder then move it to your `dsc201` RProject folder (the root folder, not inside `data`).

16. Confirm the file is in the right place by checking in RStudio's **Files** pane. You should see it in `dsc201/ds201_001_a00.Rmd`.



Step 5: Always open the `.Rproj` file

17. In your computer's file explorer, double-click `dsc201.Rproj` to start each session. This ensures your working directory and relative paths work correctly.

Knitting to PDF

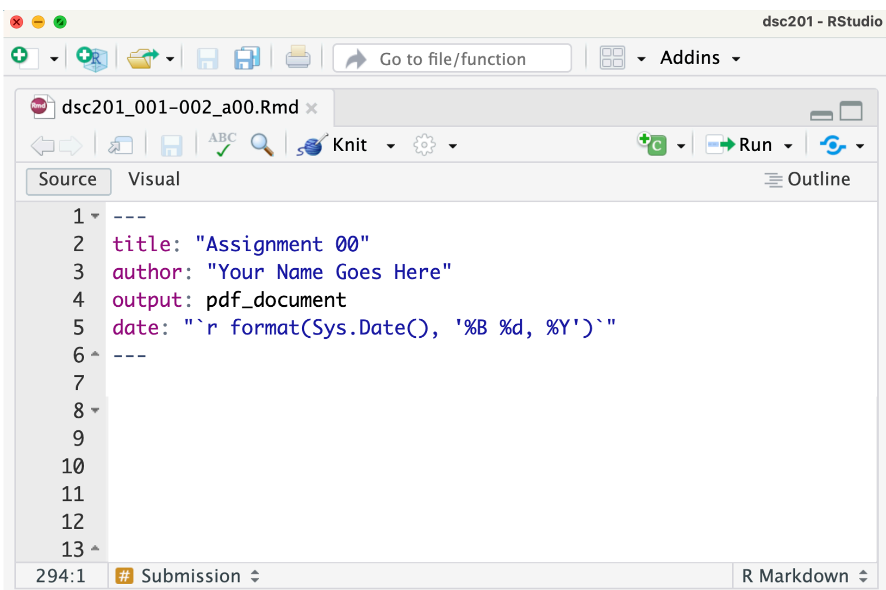
Knitting Assignment 00

In RStudio, **knitting** is the process of taking a source document written in R Markdown (`.Rmd`) and combining the text, code, and code output into a single rendered file, such as HTML, PDF, or Word. When a document is knitted, RStudio runs all code chunks, captures their output (e.g., tables, plots, and summaries), and integrates the results with the written text.

Knitting to PDF

Using the YAML Header

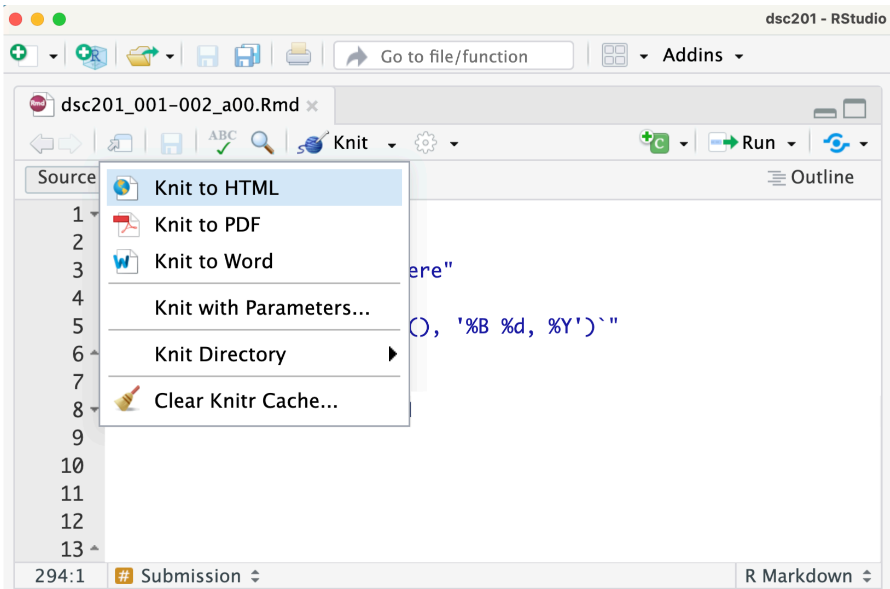
To knit directly to PDF by default, specify PDF output in the YAML header at the top of your document.



Using the Knit Menu

To manually knit a document to PDF:

1. Click the **Knit** button in RStudio.
2. Select **Knit to PDF** from the drop-down menu.



Common Knitting Errors

Missing `knitr`

If you see the following message when attempting to knit

> Required package 'knitr' is missing. Would you like to install it?

the `knitr` package has not yet been installed on your computer.

Click **Yes** and RStudio will automatically install `knitr` and any required dependencies. Once installation is complete, knitting should begin again automatically.

If the document knits successfully, no further action is required.

Missing LaTeX

If you instead see an error similar to the one below:

Warning

Error: LaTeX failed to compile RMarkdown.tex. See https://yihui.org/tinytex/r/#debugging for debugging tips. In addition: Warning message: In system2(…, stdout = if (use_file_stdout()) f1 else FALSE, stderr = f2) : ' "pdflatex" ' not found
Execution halted No LaTeX installation detected (LaTeX is required to create PDF output). You should install a LaTeX distribution for your platform: https://www.latex-project.org/get/

your computer does not have a LaTeX distribution installed. To create PDF files from R Markdown, you must install a LaTeX distribution such as TinyTeX.

Proceed to the instructions below.

Installing TinyTeX for PDF Output

To knit R Markdown documents to PDF, your computer needs a LaTeX installation. We recommend TinyTeX because it is lightweight, easy to install, and integrates well with RStudio.

TinyTeX vs. `tinytex`

TinyTeX

TinyTeX is a lightweight LaTeX distribution used to compile R Markdown documents into PDF files. It performs the actual PDF creation process.

`tinytex`

`tinytex` is an R package that installs and manages TinyTeX from within RStudio.

Steps to Set Up TinyTeX

Step 1: Install the `tinytex` Package

- Open RStudio.
- In the **Console**, run:

```r
install.packages("tinytex")
```

Note: This installs the `tinytex` package, which manages the TinyTeX installation.

Step 2: Install TinyTeX

- In the **Console**, run:

```r
tinytex::install_tinytex()
```

Note: This command downloads and installs TinyTeX on your computer and may take several minutes.

Step 3: Verify the Installation

- In the **Console**, run:

```r
tinytex::tinytex_root()
```

If the command returns a file path, TinyTeX has been installed successfully.

Example paths:

- macOS/Linux: `"/usr/local/texlive/..."`
- Windows: `"C:\\Users\\YourName\\AppData\\Roaming\\TinyTeX\\..."`

Step 4: Knit to PDF

- Open your `.Rmd` file in RStudio.
- Verify that the YAML header specifies PDF output:

```yaml
---
title: "Your Title"
author: "Your Name"
output: pdf_document
---
```

- Click **Knit**.
- Select **Knit to PDF** if prompted.

RStudio will now use TinyTeX to generate a PDF version of your document.

Tag System Overview

Purpose

This section explains the tag system used in grading and feedback for activity assignments, programming assignments, and final project milestones.

The goal of this system is to provide transparent, consistent, and instructional feedback that helps you understand how your work was evaluated and how to improve.

Tag System Overview

Assignment comments may include two kinds of tags:

- **Severity tags** describe how serious the issue is and determine the point deduction.
- **Content tags** describe what kind of issue occurred.

Tags appear at the start of comments in this format:

`[SEVERITY][CONTENT-TAG]`

For example:

`[S][CODE-REQ]`

This means the response has a major issue related to the code that was submitted to answer the question.

Severity Tags

Severity tags indicate the overall impact of an issue on correctness or completion. The severity tag determines the point deduction.

Multiple content tags may apply to a response, but only one severity level is assigned.

Content Tags

Content tags identify issues related to the completeness, correctness, clarity, interpretation, organization, or communication of a response.

The specific content tags available differ by assignment type.

Multiple Tags and One Deduction

Multiple tags may apply to the same response. However, deductions reflect the overall severity of the issue rather than treating each tag as a separate deduction.

Content tags identify the specific issues present, while the severity tag determines the point deduction.

For example, consider the following question:

> Calculate the average calories for each restaurant and identify the restaurant with the highest average calories.

A student submits:

```
ms2022 |>
  count(restaurant)
```

Possible tags:

`[S][CODE-REQ][CODE-RESULT][REQ-INCOMP]`.

- `[S]` because most required elements are missing or incorrect.
- `[CODE-REQ]` because the required computation is missing.
- `[CODE-RESULT]` because the result does not support the analysis requested.
- `[REQ-INCOMP]` because the question was not fully answered.

Although multiple content tags apply, the response receives one severity level based on the overall quality of the answer.

Activity Assignments

Grading

Activity assignments are worth 5 points total. Points are awarded based on submission timeliness and submission compliance.

- **1 point** for submission timeliness.
- **4 points** for submission requirements, which include completing all parts of the activity, following formatting requirements, and submitting all required components.

Severity Tags

Submission Timeliness

Severity Tag	Points Deducted	Points Earned	Explanation
`[T-ON]`	0.0	1.0	Submitted on time
`[T-L1]`	0.625	0.375	Submitted 1 day late
`[T-L2]`	0.775	0.225	Submitted 2 days late
`[T-N]`	1.0	0.0	Submitted more than 2 days late or not submitted

Submission Compliance

Severity Tag	Points Deducted	Points Earned	Explanation
`[C-M]`	0.0	4.0	Submission requirements met
`[C-C]`	0.5	3.5	Required component missing
`[C-S]`	2.0	2.0	Required section missing
`[C-F]`	3.0	1.0	Incorrect submission format
`[C-N]`	4.0	0.0	Not submitted

Content Tags

Assignment Compliance

Tag	Explanation
`[REQ-INCOMP]`	Required part of the question was not answered.
`[REQ-REF]`	Required reference or citation is missing.
`[REQ-MISS]`	Required element is missing.

Writing & Communication

Tag	Explanation
`[WR-FLOW]`	Response lists disconnected observations instead of explaining ideas in sentences.
`[WR-FMT]`	Excessive bolding, unnecessary headings, or inconsistent spacing distracts from readability.

Programming Assignments

Grading

Programming assignments are worth 15 points total. Points are awarded based on participation and completion of the assignment questions.

Participation [5 Points]

- **4 points** for submission timeliness.
- **1 point** for submission compliance, which includes following formatting requirements and submitting all required components.

Submission Timeliness

Severity Tag	Points Deducted	Points Earned	Explanation
`[T-ON]`	0.0	4.0	Submitted on time
`[T-L1]`	2.0	2.0	Submitted 1 day late
`[T-L2]`	3.0	1.0	Submitted 2 days late
`[T-N]`	5.0	0.0	Submitted more than 2 days late or not submitted

Submission Compliance

Severity Tag	Points Deducted	Points Earned	Explanation
`[C-M]`	0.0	1.0	Submission requirements met
`[C-N]`	0.0625	0.9375	Name missing or incorrect
`[C-D]`	0.0625	0.9375	Date missing or incorrect
`[C-S]`	0.0625	0.9375	Section missing
`[C-F]`	1.0	0.0	Incorrect submission format

Severity Tag	Points Deducted	Points Earned	Explanation

Questions [10 Points]

- **1 point** per question for a total of 10 questions.

Programming assignments include code-based questions and may also include written-response questions. Unless otherwise specified, written responses must:

- be written in third person,
- use proper grammar, punctuation, and spelling, and
- be formatted using standard font (e.g., no excessive bolding or italics).

Severity Tags

Severity Tag	Points Deducted	Points Earned	Explanation
`[C]`	0.0	1.0	The response correctly addresses the question and meets all requirements.
`[e]`	0.0625	0.9375	The work is correct, with issues limited to presentation or precision.
`[m]`	0.125	0.875	The core idea is correct, but some details are missing or unclear.
`[M]`	0.25	0.75	A key element is missing or incorrect.
`[MJ]`	0.50	0.50	The main idea or a required element is missing or incorrect.
`[S]`	0.75	0.25	An attempt was made, but most elements are missing or incorrect.
`[X]`	1.00	0.00	Missing or required work is not present.

Content Tags

Assignment Compliance

Tag	Explanation
`[REQ-INC]`	Incorrect assignment submitted.

Tag	Explanation
`[REQ-FMT]`	Notebook or file structure was altered, or an incorrect format was submitted.

Question Compliance

Tag	Explanation
`[REQ-INCOMP]`	Required part of the question was not answered.
`[REQ-REF]`	Required reference or citation is missing.
`[REQ-VIS]`	The visualization requirement was not met or properly followed.

Code & Output

Tag	Explanation
`[CODE-ERR]`	Code produces a syntax or runtime error and does not execute successfully.
`[CODE-PH]`	Placeholder or instructional comment was not removed (e.g., `# YOUR CODE GOES HERE`).
`[CODE-FMT]`	Comments and/or code run off the page or are not formatted for readability.
`[CODE-STYLE]`	Code works but does not follow required style or structure.
`[CODE-OUT]`	Entire datasets or unnecessary output are displayed when only summary output was needed.
`[CODE-MISS]`	Required output is missing, hidden, truncated, or difficult to read.
`[CODE-REQ]`	Required code, computation, or logic is missing.
`[CODE-RESULT]`	Code runs without errors, but the output does not answer the question or support the analysis.
`[CODE-EXTRA]`	Extra code is included that does not contribute to the analysis.

Visualizations

Tag	Explanation
`[VIS-CLAR]`	Visualization is cluttered, overcrowded, or difficult to interpret visually.
`[VIS-HIST]`	Histogram construction misrepresents continuous data.

Tag	Explanation
`[VIS-CHOICE]`	The visualization type is not appropriate for the structure or type of data being displayed (e.g., using a scatterplot or box plot for discrete categorical data, or using a bar chart for continuous numerical data).
`[VIS-LAB]`	The visualization is not labeled clearly or appropriately (e.g., axis labels, titles).

Writing & Communication

Tag	Explanation
`[WR-3P]`	Uses first-person language instead of third-person (e.g., "I think the graph shows…").
`[WR-OBJ]`	Uses subjective or informal language instead of objective description (e.g., "This graph is really good and interesting.").
`[WR-FLOW]`	Response lists disconnected observations instead of explaining ideas in sentences or paragraphs.
`[WR-FMT]`	Excessive bolding, unnecessary headings, or inconsistent spacing distracts from readability.
`[WR-PH]`	Instructional text such as "Replace this text with your answer" was not removed.
`[WR-REQ]`	A required written response is missing.
`[WR-CODE]`	References commands or functions in the written interpretation (e.g., "I used `group_by()` to calculate the averages.").
`[WR-QNUM]`	Includes question numbers in the written response (e.g., "Question 3:" included directly in the written interpretation).

Interpretation & Reasoning

Tag	Explanation
`[INT-REQ]`	A required explanation or interpretation component is missing.
`[INT-DESC]`	Output is reported but not interpreted (e.g., "The mean age is 42.3." with no explanation of what the value means in context).
`[INT-VAL]`	Interpretation lacks specific numerical values (e.g., "One group was higher than the other" without reporting numerical values).
`[INT-CTX]`	Results are not explained in context or why they matter.
`[INT-REL]`	Variables are discussed without explaining their relationship (e.g., mentions two variables separately without explaining how they relate).

Tag	Explanation
[INT-INF]	Interprets information that is not shown in the output (e.g., claims a trend or relationship that is not supported by the displayed output).
[INT-COR]	Uses correlation without verifying that it is an appropriate measure of the relationship.
[INT-CA]	Uses causal language such as *affects*, *impacts*, or *causes* (e.g., "More exercise causes higher happiness levels") based only on exploratory analysis.

Final Project Milestones

Final Project Milestones

Milestone 1

Dataset Approval Request [15 Points]

In this milestone, you will identify a dataset and demonstrate that it satisfies the project requirements. Detailed submission instructions are available in Moodle.

Grading

Dataset Approval Request is worth 15 points total. Points are awarded based on submission timeliness and dataset approval.

Submission Timeliness [5 Points]

Severity Tag	Points Deducted	Points Earned	Explanation
`[T-ON]`	0.0	5.0	Submitted on time
`[T-L1]`	2.0	3.0	Submitted 1 day late
`[T-L2]`	3.5	1.5	Submitted 2 days late
`[T-L3]`	4.0	1.0	Submitted more than 2 days late
`[T-N]`	5.0	0.0	Not submitted

Dataset Approval [10 Points]

Severity Tag	Points Deducted	Points Earned	Explanation
`[D-APP]`	0.0	10.0	The dataset meets all requirements and all required submission elements are present. **Dataset approved.**

Severity Tag	Points Deducted	Points Earned	Explanation
[D-MF]	2.0	8.0	The dataset meets most requirements, but one or two issues must be addressed before approval. **Dataset not yet approved.**
[D-PF]	5.0	5.0	Several requirements are unmet or unclear. Significant revision or selection of an alternate dataset is required. **Dataset not yet approved.**
[D-SM]	8.0	2.0	Many requirements are unmet or violated. The dataset is not viable in its current form and should be replaced. **Dataset not yet approved.**

Content Tags

Dataset Structure & Size Issues

Tag	Explanation
[D-FTYPE]	File type issue.
[D-FSIZE]	File size issue.
[D-RANGE]	The dataset does not meet the 500–10,000 row requirement.
[D-VAR]	The dataset does not include at least 10 variables.
[D-CAT]	Fewer than 2 categorical variables are present.
[D-NUM]	Fewer than 2 numerical variables are present.

Dataset Suitability Issues

Tag	Explanation
[D-TS]	The dataset is primarily time-series data.
[D-REP]	The dataset contains repeated measurements of the same subject.
[D-PII]	Personally identifiable information (PII) is present in the dataset.
[D-OVR]	The dataset is overly common or has extensive publicly available analyses.
[D-AN]	The dataset is unlikely to support meaningful analysis using the methods covered in this course.

Dataset Access & Documentation Issues

Tag	Explanation
[D-UO]	The unit of observation is unclear or incorrectly described.
[D-LINK]	The dataset link is missing, broken, or requires a personal login.

Dataset Justification & Research Question Issues

Tag	Explanation
[J-FIT]	The justification does not clearly explain why the dataset fits the project goals.
[J-REQ]	The justification does not explain how the dataset meets the project requirements.
[J-RQ]	Example research questions are missing or unclear.

Datasheet (15 Points)

In this milestone, you will document the variables, structure, and context of your dataset. Detailed submission instructions are available in Moodle.

Submission Timeliness

Submission Compliance

Content Tags

Milestone 2

Data Dive Using R and Python (15 Points Each)

In these milestones, you will use R and Python to explore your dataset, identify patterns and potential issues, and generate insights that will help inform your final analysis. Detailed submission instructions are available in Moodle.

See Programming Assignments for grading criteria.

Milestone 3

Analysis Plan & Research Questions (15 Points)

In this milestone, you will develop research questions and identify the variables that will be used in your analysis. Detailed submission instructions are available in Moodle.

Submission Timeliness

Submission Compliance

Content Tags

Data and Analysis Notebook Setup (15 Points)

Submission Timeliness

Submission Compliance

Content Tags

Milestone 4

Data Analysis & Storytelling (25 Points)

Severity Scale for 3-Point Components

Severity Scale for 2-Point Components

Content Tags